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Course Code: CMP 412

Course Title: Computer Network and Communication

Question:

Write a code to show the communication between multiple clients and the server.

Server.py

```
import socket
from _thread import start_new_thread

def function(con):
    while True:
        data = con.recv(1024).decode()
        if not data:
            break
        con.sendall(data.upper().encode())

    con.close()

ss = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)

ip = socket.gethostname()
port = 9000

ss.bind((ip,port))
ss.listen(5)

print("Server start")

while True:
    con,addr = ss.accept()
```

```
start_new_thread(function, (con,))
```

Client_1.py

```
import socket
cs = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)

ip=socket.gethostname()
port = 9000

cs.connect((ip,port))

while True:
    msg = input("Enter a message or enter by to quit---> ")
    if msg.strip().lower()=="bye":
        break
    cs.sendall(msg.encode())

    data=cs.recv(1024).decode()
    print("server response--> ", data)
```

Client_2.py

```
import socket
cs = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)

ip=socket.gethostname()
port = 9000

cs.connect((ip,port))

while True:
    msg = input("Enter a message or enter by to quit---> ")
    if msg.strip().lower()=="bye":
        break
    cs.sendall(msg.encode())

    data=cs.recv(1024).decode()
    print("server response--> ", data)
```

Client_3.py

```
import socket
cs = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)

ip=socket.gethostname()
port = 9000
```

```

cs.connect((ip,port))

while True:
    msg = input("Enter a message or enter by to quit---> ")
    if msg.strip().lower()=="bye":
        break
    cs.sendall(msg.encode())

    data=cs.recv(1024).decode()
    print("server response--> ", data)

```

Output

The screenshot displays four terminal windows in VS Code, all running the same command prompt at the directory `PS C:\Users\Pelumi of God\Documents\400-Level-First-Semester\CHP 412\server to client assignment>`.

- Terminal 1:** Shows the server starting with the command `py server.py`. The output is `Server start` followed by a blank line.
- Terminal 2:** Shows a client sending `hello` and receiving `server response--> HELLO`. The client then enters `bye` to quit.
- Terminal 3:** Shows a client sending `hi` and receiving `server response--> HI`. The client then enters `bye` to quit.
- Terminal 4:** Shows a client sending `hello` and receiving `server response--> HELLO`. The client then enters `bye` to quit.

On the right side of the interface, there is a file explorer showing the files `py`, `pwsh`, and `pwsh` in the current directory.